# Admissible Non-Injective Transitions as the Primitive of Physical Description

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**Abstract**

Quantum mechanics is the effective theory of physically real unresolved admissible structures and their resolution. This paper establishes the minimal axiomatic foundation from which this claim follows: physical structure is derived from a single primitive, the local structure of admissible transitions between observable states. Four axioms govern this structure: local projective admissibility (A1), structural non-injectivity (A2), admissibility as non-premature selection (A3), and projection locking (A4). From A1 and A2 alone, we derive irreversibility, the arrow of time, and the existence of structural fluctuations, without any additional postulate. From A3, we derive the proto-state as a physically real unresolved configuration shared between two successive states, and show that admissibility forces the retention of phase coherence. We further prove that A3 forbids any admissible factorisation of the fibre, which forces a non-trivial commutator between conjugate generators; combined with minimality (A1) and Born–Infeld parity, this identifies the symmetry group of the admissible fibre as $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime $q$, and the fibre itself as the Weil representation $V_\rho$ — closing the identification $F_n \simeq V_\rho$ as a theorem rather than a remark. From A4, we derive the discrete character of quantum transitions as a consequence of the interaction between continuous Born–Infeld saturation and the discrete shell structure of the observable space. This paper provides the axiomatic foundation cited by the companion papers [8, 1, 13] and serves as the formal basis for the derivations carried out therein.

**Keywords.** Non-injective projection; Emergence; Projective structure; Admissibility; Born–Infeld constraint; Information loss; Projection entropy; Quantum emergence; Phase coherence; SU(2) structure; Gauge emergence; Fiber structure; Projective temporal ordering; Irreversibility.

## Contents

# 1 Introduction

The Cosmochrony programme [4] aims to derive the structure of physical description from a minimal relational foundation, without postulating a background spacetime, a Hilbert space, or an external substrate. The only primitive is the local structure of admissible transitions between observable states.

The central question driving the framework is:

> Given an observable state $O_{n-1}$, what states are accessible from it?

The answer to this question — encoded in the admissibility structure — is sufficient to derive irreversibility, temporal order, wave-like propagation, phase coherence, and the discrete character of quantum transitions. No substrate, no background, and no quantum postulate is required as input.

### Relation to companion papers

This paper provides the axiomatic foundation for a chain of companion results. The companion paper [1] establishes that Bell-type factorisation fails generically under non-injective projection (A2), without invoking nonlocal dynamics. The companion paper [13] proves that non-injectivity is a structural necessity of any genuinely emergent description. The companion paper [8] derives phase coherence, the singlet correlator, the Tsirelson bound, and the Born rule as structural consequences of admissibility — using Axioms A1–A4 as stated here, combined with the concrete spectral results of the O-series. The white paper [4] develops the full phenomenological programme; the present axiomatic reformulation is a foundational companion to it, not yet integrated into its main text.

### Organisation

Section 2 states the four axioms. Section 3 clarifies the role of $\Pi$ and the index $n$. Section 4 derives the structural consequences of non-injectivity: irreversibility, novelty, temporal order, and the projective incompleteness principle. Section 5 derives the proto-state, phase, coherence, and the Heisenberg structure of the admissible fibre from the admissibility principle. Section 5.7 derives projection locking and the discrete character of quantum transitions. Section 6 treats entanglement and Bell non-applicability. Section 7 summarises what is derived from what, and identifies the open directions.

## 2 The Four Axioms

We state the four axioms from which the framework proceeds. They are not independent postulates chosen for convenience — each captures a structural feature that is either necessary for genuine emergence [13] or forced by the requirement of bounded effective descriptions [3].

**Definition 2.1** (Admissible transition). *A transition from observable state $O_{n-1}$ to observable state $O_n$ is* admissible *if it satisfies Axioms A1–A4 below.*

**Remark 2.2** (Admissibility is relational, not absolute). *Admissibility is never an intrinsic predicate of an observable state $O_n$ taken in isolation. It is a relational constraint on projective transitions: a state $O_n$ is admissible only relative to a preceding state $O_{n-1}$, whose accumulated constraints determine the local admissible neighbourhood. We write $O_n \in \mathcal{A}(O_{n-1})$ to make this explicit, where*

$$\mathcal{A}(O_{n-1}) \;:=\; \big\{\, O_n \mid \textit{the transition } O_{n-1} \to O_n \textit{ satisfies A1–A4} \,\big\} \tag{1}$$

*is the* admissible neighbourhood from $O_{n-1}$*. The phrasing "$O_n$ is admissible" should always be read as shorthand for $O_n \in \mathcal{A}(O_{n-1})$: admissibility is always* from-here *admissibility, conditioned on the already-realized projective position. There is no fixed space of possible observable states: there is a field of local admissible neighbourhoods, each determined by the point already reached.*

**A1 — Local projective admissibility.** For any observable state $O_{n-1}$, there exists a non-empty set $F_n$ of admissible successor directions: configurations of $O_{n-1}$ compatible with a transition to a stable next state. The set $F_n$ is defined by internal structural constraints (Born–Infeld bound, spectral envelope, symmetry requirements) and does not presuppose an ambient configuration space.

**A2 — Structural non-injectivity.** Distinct admissible directions from $O_{n-1}$ may lead to the same resolved successor state $O_n$. The map $\Pi_n$ encoding this is generically non-injective.

**A3 — Admissibility as non-premature selection.** An admissible transition preserves the full multiplicity of open directions until projection locking occurs. It does not select among them prematurely — that is, before the Born–Infeld saturation threshold is reached [3].

**A4 — Projection locking.** Resolution occurs only when continuous Born–Infeld saturation meets the discrete shell support of the observable space. The transition from admissible to admitted is structurally discrete.

**Remark 2.3** (Independence of the axioms). *A1 asserts existence of admissible directions. A2 asserts their generic non-injectivity. A3 constrains which transitions are admissible. A4 governs when resolution occurs. The four axioms are logically independent: none follows from the others. Their joint necessity is argued in Section 3 and in the companion papers [13, 3].*

**Remark 2.4** (Status of Born–Infeld in A3 and A4). *The Born–Infeld saturation threshold appearing in A3 and A4 is not an additional postulate: it is the unique bounded completion of the effective action compatible with bounded projective flux, established in [3]. A3 and A4 therefore do not introduce new structure beyond A1–A2 and the Born–Infeld companion.*

**Remark 2.5** (What projection locking selects). *A4 resolves the transition to a unique observable state $O_n$ in the image of $\Pi_n$. It does not select a unique underlying configuration in $F_n$.*

*The multiplicity of admissible configurations in the fibre $F_n = \Pi_n^{-1}(O_n)$ is not eliminated by projection locking: distinct configurations $f, f' \in F_n$ with $\Pi_n(f) = \Pi_n(f') = O_n$ remain structurally present after locking, but become observationally inaccessible. The non-injectivity of $\Pi_n$ therefore persists at every resolution step.*

*This distinction — observable uniqueness without configurational uniqueness — prevents Bell-type factorisation [1] and forces the coherent, non-factorisable fibre structure derived in Section 5.4. More generally, distinct interactions $I_1 \neq I_2$ induce distinct admissible sub-fibres $F_n^{(I_1)} \neq F_n^{(I_2)}$ and hence distinct observable outcomes, without any element of $F_n$ having been selected. This is the structural origin of contextuality: measurement context determines which sub-fibre is accessed, not which configuration is realised. This contextuality is* synchronic*: it concerns the choice of sub-fibre at a fixed transition step n. The admissible neighbourhood $\mathcal{A}(O_{n-1})$ (Remark 2.2) introduces a second,* diachronic *form of contextuality: the set of accessible observable states at step n is itself conditioned on the previously realized state $O_{n-1}$. The context of a projection is therefore not merely the measurement algebra chosen at step n, but also the already-realized projective position from which the next state may be reached. Standard contextuality frameworks (Kochen–Specker, sheaf-theoretic) capture synchronic contextuality; the present framework extends this to diachronic contextuality, in which the history of realized projections is constitutive of what is admissible next.*

**Remark 2.6** (Projective Observability Principle). *Axioms A1–A4 jointly imply the following criterion, which identifies what is physical in the framework.*

Physical observables coincide with the invariants of the projection $\Pi_n$: any quantity that depends on the choice of pre-image within $\Pi_n^{-1}(O_n)$ is unphysical by construction.

*This is not an additional axiom but a consequence of A2 (which forces non-trivial fibres) and A4 (which identifies* $O_n$ *with* $\operatorname{Im}\Pi_n$ *exactly): since* $\Pi_n$ *is the only structure that relates the configurational level to the observable level, any quantity invisible to* $\Pi_n$ *is invisible to physics. What the projection erases — intra-fibre distinctions — is unphysical; what it preserves — the invariants of the fibre structure — constitutes the physical content of the theory. In particular, the symmetry group of a fibre* $\Pi_n^{-1}(O_n)$ *is physical precisely because it acts within the fibre while leaving* $O_n$ *unchanged: this is the structural origin of gauge symmetry, developed in the companion papers [9, 26]. A detailed formulation in terms of the five-level hierarchy is given in Remark 3.5.*

# 3 The Primitive and the Role of $\Pi$

## 3.1 The primitive

The only primitive of this framework is the pair $(O_{n-1}, F_n)$: an observable state and the set of admissible directions it admits. There is no external substrate, no hidden carrier, no background geometry. The fibre $F_n$ is not defined in an abstract ambient space — it is induced by $O_{n-1}$ itself through the admissibility constraints of A1–A3.

The resulting evolution is recursive: once $O_n$ is realized, it immediately becomes the new departure point, generating the next admissible neighbourhood $\mathcal{A}(O_n)$ from which $O_{n+1}$ may be reached:

$$O_{n-1} \longrightarrow O_n \longrightarrow \mathcal{A}(O_n) \longrightarrow O_{n+1}. \tag{2}$$

The realized state is simultaneously a destination and the generator of the next set of constraints. No external agent resets the context between steps: the admissible neighbourhood at each stage is determined entirely by the state just reached.

## 3.2 The projection $\Pi_n$ as descriptive abstraction

The notation $\Pi_n$ is a compact encoding of the local admissibility structure. It encodes three things simultaneously:

(i) the admissible directions from $O_{n-1}$ — the fibre $F_n$ (A1, A2);

(ii) the Born–Infeld saturation bound — the limit on accessible amplitudes (A3);

(iii) projection locking — the constraint that resolution falls on an available shell (A4).

$\Pi_n$ is not a physical entity and not a dynamical map. It is a structural relation encoding the admissible identification between distinguishable configurations (elements of $F_n$) and indistinguishable observable outcomes (elements of $O_n$). Saying "$\Pi_n$ is non-injective" means that multiple directions from $O_{n-1}$ can lead to the same $O_n$ — a fact about the admissible transition structure, not a property of an external map. Non-injectivity does not originate from an underlying substrate, but from the structure of admissible configurations and their interaction-dependent realisation map.

**Remark 3.1** (The threefold role of $\Pi_n$)**.** *The structural consequences established below in Sections 4 and 4.4 reveal that* $\Pi_n$ *plays three distinct roles simultaneously, each irreducible to the others.*

*(i)* Admissibility filter. $\Pi_n$ *selects, from the structurally compatible continuations* $\Omega_n^{(c)}$*, those directions satisfying the Born–Infeld and spectral constraints of A1–A3. What it retains constitutes the observable* $O_n$*; what it discards is the non-trivial kernel whose existence is guaranteed by Corollary 4.6.*

*(ii)* Generator of temporal order. *The oriented sequence* $\Pi_0, \Pi_1, \Pi_2, \ldots$ *carries the partial order established in Theorem 4.2. Each* $\Pi_n$ *is a node in the admissibility graph (Remark 4.4); the index* $n$ *acquires its temporal meaning from this orientation, not from an external parameter. The dynamics of the framework is the oriented topology of this graph — not a law superimposed on a pre-existing time.*

*(iii)* Revelation, not creation. $\Pi_n$ *does not produce new structure in* $\chi$*. It selects, from the admissible spectral modes of the Weil representation already present in* $\Omega_n^{(c)}$*, those that are observationally accessible at depth* $n$*. The BFS depth controls how much of the structure of* $\chi$ *has been revealed; increasing* $n$ *does not modify* $\chi$ *but resolves finer spectral content — as a deeper lens resolves detail that was always present in the object, not detail that the lens creates.*

*These three roles are inseparable: the structural constraints A1–A3 and Born–Infeld parity that force the admissibility filter simultaneously generate the temporal order via Theorem 4.2 and guarantee the incompleteness of every resolution via Corollary 4.6. The non-commutativity* $[X, \sigma(X)] \neq 0$ *at the root of the incompleteness (Theorem 5.7) is therefore not merely a technical property of the fibre: it is the algebraic source of all three roles at once.*

### 3.3 The index $n$

The index $n$ does not represent a pre-existing time parameter. It counts transitions: the number of times a state has been succeeded by another under the admissibility constraints. Its temporal meaning is derived in Section 4 — it is not assumed.

### 3.4 Where non-injectivity lives

A natural question arises: where does the non-injectivity of $\Pi_n$ come from, given that no underlying substrate is postulated? The answer is precise.

Non-injectivity is a property of the interaction-dependent realisation map: the map from the space $F_n$ of structurally compatible configurations to the space $O_n$ of observable outcomes. When two distinct admissible configurations $f, f' \in F_n$ give rise to the same observable outcome $o \in O_n$, it is not because they are "the same at a deeper level" — it is because the interaction that realises the transition cannot distinguish them within $O_n$. The limit on distinguishability is set by the Born–Infeld capacity constraint (A3): the observable space $O_n$ has finite projective capacity, and configurations that differ only below this threshold are identified.

Three properties of this realisation map are essential:

(i) *Interaction-dependence*: $\Pi_n$ encodes the interaction; different interactions define different realisation maps and different fibres $F_n$.

(ii) *Internal definition of admissibles*: $F_n$ is defined by internal constraints (BI bound, spectral envelope, symmetry) — not by reference to an ambient substrate.

(iii) *Minimal structure of* $F_n$: the admissibility constraints force $F_n$ to carry a specific algebraic structure — SU(2) in the spin-$\frac{1}{2}$ sector (O23) — which is derived from the constraints, not assumed.

The non-injectivity is therefore neither a property of a hidden substrate nor an arbitrary assumption: it is a structural consequence of finite projective capacity applied to an internally defined space of admissible configurations.

**Remark 3.2** (The admissible fibre as equivalence class under projection)**.** *A useful image for the non-injectivity of* $\Pi_n$ *is that of shadows cast by a fixed projection setup. A given shadow does not correspond to a single object: it corresponds to the set of all objects whose projection yields that shadow — the admissible fibre* $F_n = \Pi_n^{-1}(O_n)$*.*

*The directionality is essential. The fibre is not the collection of all shadows produced by a given object under varying projections (one configuration, many images). It is the collection of all configurations that produce the same observable under a fixed, non-injective* $\Pi_n$ *(many configurations, one observable). This many-to-one structure is the source of irreversibility (Section 4) and of the non-factorisability of the proto-state (Section 5.4).*

**Remark 3.3** (Connection to the O-series: $F_n$ and $V_\rho$)**.** *In the concrete spectral realisation of the O-series, the admissible fibre* $F_n$ *is realised as a spectral sector* $V_\rho$ *of the Heisenberg Cayley graph* $\mathrm{Cay}(\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$*. The sectors* $V_\rho$ *are not pre-existing configuration spaces: they are the structures induced by the admissibility constraints — the Born–Infeld bound, the spectral envelope, and the symmetry requirements of A1. In this sense,* $V_\rho$ *is the concrete realisation of* $F_n$*, not an ambient space in which* $F_n$ *is embedded:*

$$F_n \;\simeq\; V_\rho. \tag{3}$$

*This identification, stated here as a structural connection to the O-series, is established as a theorem in Section 5.4: the axioms A1–A3 and Born–Infeld parity together force the symmetry group of* $F_n$ *to be* $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$*, and the Stone–von Neumann theorem then identifies* $F_n \simeq V_\rho$ *uniquely.*

**Remark 3.4** (The two levels of $\Pi_n$: kernel and image (O24))**.** *The non-injective projection $\Pi_n$ operates at two structurally distinct levels, established in O24 [22]:*

*(i)* Kernel level *(*$\ker \Pi_n$*): governs microscopic multiplicity and fluctuation amplitude. Multiple configurations in $\Omega_n^{(c)}$ may project to the same observable outcome; their differences are invisible to admissible observables and appear as projection residuals. The dependence on $O_{n-1}$ — via the baseline $V_{n-1}^{(c)}$ — enters at this level: it determines which directions in $\Omega_n^{(c)}$ are new relative to the already realized subspace.*

*(ii)* Image level *(*$\operatorname{Im} \Pi_n \cap \mathrm{Ntrl}$*): governs the observable structure. Its real rank is rigidly fixed to 3 by the quaternionic maximality of* $\operatorname{Im} \mathbb{H}$ *(O23, Theorem 3.2) and is independent of the cardinality of* $\ker \Pi_n$*.*

*The Verticality Lemma of O24 (Lemma 3.1) establishes that every Born–Infeld-admissible symmetry acts vertically: it maps each fibre $\Pi_n^{-1}(o)$ to itself and does not increase the rank of the image. The structural consequence is:*

Non-injectivity affects degeneracy, not structure.

*Enlarging* $\ker \Pi_n$ *— admitting more configurations in each fibre — increases the amplitude of quantum fluctuations but does not alter the observable rank, the cascade exponent $\delta_{\mathrm{pair}}$, or any other structural invariant of the admissible sector. The two levels of $\Pi_n$ are therefore decoupled: kernel structure is physically relevant for fluctuations; image structure is physically relevant for observables.*

**Remark 3.5** (The observable space principle: $\mathcal{O} = \operatorname{Im} \Pi$)**.** *A direct consequence of the five-level hierarchy is that admissible observables are functions on*

$$\mathcal{O} \;:=\; \operatorname{Im} \Pi \;=\; \bigsqcup_n O_n, \tag{4}$$

*not on the pre-image space $\Omega_n^{(c)}$. Any structural difference between two configurations $f, f' \in \Pi_n^{-1}(O_n)$ that share the same image under $\Pi_n$ is invisible to every admissible observable by definition: it is a difference* within *the fibre, which the projection identifies as observationally indistinguishable. This is not an additional assumption but an immediate consequence of Definition 3.6 and Level 5: $O_n$ is defined as $\operatorname{Im} \Pi_n$, and admissibility is a property of $\Pi_n$ acting on $\Omega_n^{(c)}$, not of $\Omega_n^{(c)}$ itself. Intra-fibre variability is therefore structurally unobservable; it constitutes the degree of freedom whose projected manifestation appears as quantum indeterminacy in the effective description. In particular, any admissible quantity that depends on the choice of representative in $\Pi_n^{-1}(O_n)$ must agree for all representatives in the same fibre class: this is the structural basis for the uniformity of admissible spectral invariants across the fibre.*

The abstract structure of Section 3 has a concrete realisation in the O-series that answers precisely where $\Omega_n$ lives and how it is constructed. The realisation proceeds through five levels, each induced by the previous one.

**Level 1 — The relational group $G_q$.** The combinatorial substrate of the BFS exploration is the discrete Heisenberg group

$$G_q \;=\; \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}), \qquad |G_q| = q^3, \tag{5}$$

equipped with the standard symmetric generating set $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$. $G_q$ is not a physical space: it is a relational structure, encoding how configurations relate to one another via the generators. The BFS explores $G_q$ in shells $S_n = B_n \setminus B_{n-1}$, where $B_n = \{g \in G_q : d(e,g) \le n\}$. The shell structure $\{S_n\}$ is the concrete realisation of the transition-counting index $n$ of Section 3.

**Level 2 — The Weil representation $\rho_c$ and its ambient space.** For each non-zero character $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, the discrete Heisenberg group $G_q$ admits an irreducible $q$-dimensional Weil representation [6]:

$$\rho_c : G_q \;\longrightarrow\; \mathrm{GL}(\mathbb{C}^q), \qquad (\rho_c(a,b,\gamma)\, f)(x) \;=\; e^{2\pi i c(\gamma + bx)/q}\, f(x-a), \tag{6}$$

acting on $L^2(\mathbb{Z}/q\mathbb{Z}) \cong \mathbb{C}^q$. The ambient space $\mathbb{C}^q$ is not a physical configuration space presupposed by the theory: it is the natural carrier of the irreducible representation of $G_q$ of central character $c$. It is induced by $G_q$ and $c$, not by any external assumption.

**Level 3 — The orbit span $\Omega_n$ as emerging subspace.** Given the initial vector $v_0 \in \mathbb{C}^q$ and the Weil action $\rho_c$, the orbit span at BFS depth $n$ is:

$$\Omega_n^{(c)} \;:=\; \mathrm{span}\big\{\rho_c(g)\, v_0 : g \in B_n\big\} \;\subseteq\; \mathbb{C}^q. \tag{7}$$

By construction, $\Omega_n^{(c)}$ is a minimal extension of the previously realized subspace:

$$V_{n-1}^{(c)} \;\subseteq\; \Omega_n^{(c)} \;\subseteq\; \mathbb{C}^q, \tag{8}$$

where $V_{n-1}^{(c)} = \mathrm{GS}(B_{n-1})$ is the Gram–Schmidt span accumulated over the preceding shells. $\Omega_n^{(c)}$ is not defined by $O_{n-1}$, but is constrained by the already realized subspace $V_{n-1}^{(c)}$: the directions in $\Omega_n^{(c)} \setminus V_{n-1}^{(c)}$ are the candidates for new observable structure at depth $n$.

**Definition 3.6** (Structurally compatible continuations)**.** $\Omega_n^{(c)}$ *is the space of* structurally compatible continuations *from* $O_{n-1}$*: it contains every direction reachable by the relational structure of* $G_q$ *via* $\rho_c$*, without any admissibility filter applied. Admissibility is* not *a property of* $\Omega_n^{(c)}$*; it is a property of* $\Pi_n$ *acting on* $\Omega_n^{(c)}$*.*

Three properties follow immediately:

(i) *Emergence*: $\Omega_n^{(c)}$ is not given a priori — it is built incrementally as the BFS explores $G_q$.

(ii) *Indirect conditioning on $O_{n-1}$*: $\Omega_n^{(c)}$ is defined by the group action alone — the definition (7) refers only to $B_n$ and $\rho_c$. The dependence on $O_{n-1}$ is indirect: $O_{n-1}$ determines the Gram–Schmidt span $\mathrm{GS}_{n-1}$ already accumulated over $B_{n-1}$, which acts as the baseline against which new directions in $\Omega_n^{(c)}$ are measured. This baseline is used by $\Pi_n$ (not by $\Omega_n^{(c)}$ itself) to identify genuinely new directions, counted by $\sigma_c(n) = \delta r_n/|S_n|$ [6]. In summary: $\Omega_n^{(c)}$ is determined by $(G_q, \rho_c, B_n)$; the role of $O_{n-1}$ enters only through $\Pi_n$.

(iii) *Pre-admissibility character*: $\Omega_n^{(c)}$ contains all directions generated by the group action of $G_q$ up to depth $n$ — including directions that $\Pi_n$ will subsequently reject as inadmissible. The admissibility filter is entirely in $\Pi_n$, not in $\Omega_n^{(c)}$.

**Level 4 — The admissibility filter $\Pi_n$.** The projection $\Pi_n$ acts on $\Omega_n^{(c)}$ via the BI admissibility constraint $A_n \leq c_\chi/\sqrt{\lambda_n}$ and the Gram–Schmidt orthogonalisation [6, 18]. It selects, from all directions in $\Omega_n^{(c)}$, those compatible with the admissibility constraints. $\Pi_n$ encodes the laws revealed by the O-series — Born–Infeld bound, parity involution, projection locking — but does not define $\Omega_n^{(c)}$.

**Level 5 — The observable outcome $O_n$.** The observable state $O_n$ is the image of $\Pi_n$ applied to the admissible part of $\Omega_n^{(c)}$. It is a discrete, shell-indexed element of the observable space [18]:

$$O = \bigsqcup_{n\in\mathbb{N}_q} O_n. \tag{9}$$

**Remark 3.7** (Summary: the five-level hierarchy)**.** *The five levels constitute a single construction with no external inputs:*

$$G_q \xrightarrow{\rho_c} \mathbb{C}^q \xrightarrow{\text{BFS}} \Omega_n^{(c)} \xrightarrow{\Pi_n} O_n, \qquad V_{n-1}^{(c)} \subseteq \Omega_n^{(c)} \subseteq \mathbb{C}^q. \tag{10}$$

*$G_q$ is the relational structure. $\mathbb{C}^q$ is its representation carrier. $\Omega_n^{(c)}$ is the emergent space of structurally compatible continuations (not yet filtered by admissibility), constrained from below by the already realized subspace $V_{n-1}^{(c)}$. $\Pi_n$ is the admissibility filter. $O_n$ is the resolved observable state. No level presupposes the next: each is induced by the preceding structure and the admissibility constraints. This is the concrete answer to the question of where*

*$\Omega_n$ lives: in $\mathbb{C}^q$, itself induced by $G_q$ — not in an external configuration space.*

**Remark 3.8** (The conjugate pair and $\Omega_n$ without preferred direction)**.** *The parity involution $c \leftrightarrow q-c$ (O18) acts on $\Omega_n^{(c)}$ via complex conjugation [7]: $\Omega_n^{(q-c)} = \overline{\Omega_n^{(c)}}$. The minimal admissible fibre is therefore the pair $\{V_n^{(c)}, V_n^{(q-c)}\} = \{V_n^{(c)}, \overline{V_n^{(c)}}\}$, with no preferred orientation — consistent with the admissibility principle of A3.*

### 3.5 From $\mathbb{C}^q$ to continuous spacetime: the large-$q$ limit

The five-level hierarchy of Section 3.4 is formulated for finite prime $q$. Physical spacetime is continuous. This subsection identifies the structural bridge between the two, without claiming to complete the derivation.

**The discrete structure at finite $q$.** At finite $q$, the observable space $O = \bigsqcup_{n \in \mathbb{N}_q} O_n$ is discrete and finite. The ambient representation space $\mathbb{C}^q$ has dimension $q$. Physical quantities are functions on $\mathbb{Z}/q\mathbb{Z}$.

**The large-$q$ limit.** As $q \to \infty$ with the prime structure retained, three things happen simultaneously:

(i) *The representation space $\mathbb{C}^q$ approaches $L^2(\mathbb{R})$.* Functions on $\mathbb{Z}/q\mathbb{Z}$ with fixed Fourier support become functions on $\mathbb{R}$ in the Pontryagin dual limit. The Weil representation $\rho_c$ of $G_q$ approaches the metaplectic representation of the real Heisenberg group $\mathrm{Heis}_3(\mathbb{R})$.

(ii) *The BFS shell structure approaches a continuous foliation.* The polynomial ball growth $|B_n| \sim c\, n^4$ (Bass–Guivarc'h, homogeneous dimension $D = 4$) induces, in the $q \to \infty$ limit, a continuous layering of the configuration space compatible with a four-dimensional effective geometry [4].

(iii) *The discrete admissibility filter $\Pi_n$ approaches a differential constraint.* The Born–Infeld admissibility bound $A_n \leq c_\chi/\sqrt{\lambda_n}$ becomes, in the continuum limit, the Born–Infeld Lagrangian constraint on effective field configurations [3].

The continuum spacetime observable is therefore not a separate postulate: it is the large-$q$ limit of the discrete observable space $O$, with the four-dimensional structure inherited from the homogeneous dimension of $G_q$.

**Remark 3.9** (Status of the continuum limit)**.** *The convergence $\mathbb{C}^q \to L^2(\mathbb{R})$ and the emergence of continuous spacetime geometry from the BFS shell structure are established qualitatively in [4]. The companion paper [11] initiates the rigorous treatment of this limit, formulating it in the category of*

*Hilbert spaces and operator algebras and decomposing it into three distinct convergences: a Hilbert inductive limit (T1), a representational convergence of Weil generators to metaplectic generators (T2), and a Mosco convergence of admissibility forms to a second-order operator $L_\Pi$ on $L^2(\mathbb{R})$ (T3). The continuum limit is reduced to a single spectral tightness condition (Conjecture C of [11]), which is confirmed quantitatively across all tested primes: the spectral tail mass of admissible test vectors satisfies $E_q(q/3)/\|f_q\|^2 < 2.1 \times 10^{-5}$ uniformly, with values decreasing toward machine precision as q grows. The embedding hypothesis [H1] is proved in the low-frequency regime relevant to admissible sequences; the Weil-block compatibility condition and the identification of the effective metric from the symbol of $L_\Pi$ are established in the companion paper Q5b [2]. The present remark identifies the mechanism; [11] establishes the operator-theoretic framework and the numerical evidence.*

**Remark 3.10** (Why four dimensions)**.** *The homogeneous dimension $D = 4$ of the Heisenberg group $G_q$ (Bass–Guivarc'h theorem: $|B_n| \sim n^4$) is not chosen to match the observed four dimensions of spacetime. It is a structural consequence of the nilpotency class and rank of* $\mathrm{Heis}_3$*: the group has three generators with one non-trivial commutator, giving $D = 2 \cdot \mathrm{rank} + \mathrm{class} = 4$. The coincidence with the observed dimension of spacetime is therefore a structural prediction of the framework, not a parameter fit.*

# 4 Structural Consequences of Non-Injectivity

Non-injectivity (A2) and the existence of distinct successor states together imply three structural facts whose sources are distinct.

## 4.1 Irreversibility

**Proposition 4.1** (Irreversibility from non-injectivity)**.** *Under A2, every admissible transition involves an irreversible loss of structural information.*

*Proof.* Since multiple directions from $O_{n-1}$ map to the same $O_n$ under $\Pi_n$, the information distinguishing these directions is absent from $O_n$. No subsequent transition $\Pi_{n'}$ for $n' > n$ can recover this information: the map $\Pi_n$ has positive projection entropy $S_{\Pi_n} > 0$ [13], and no admissible transition can reduce $S_\Pi$ below zero. The loss is therefore structural and permanent. □

## 4.2 Novelty and structural fluctuations

The existence of a distinct successor state $O_n \neq O_{n-1}$ defines a structural residue:

$$\Delta_n \ := \ \mathrm{Res}(O_{n-1}, O_n; \Pi_n), \tag{11}$$

the measure of how much $O_n$ is irreducible to a continuation of $O_{n-1}$ under $\Pi_n$. That $\Delta_n$ is non-trivial is not an additional assumption — it is the condition under which we index $O_n$ as a distinct state. These residues are what physical descriptions call fluctuations or quantum noise.

## 4.3 Temporal order

**Theorem 4.2** (Temporal order from A2)**.** *Under A2, the chain of observable states $\{O_n\}_{n\geq 0}$ carries a canonical oriented, non-trivial partial order.*

*Proof.* Irreversibility (Proposition 4.1) establishes an asymmetric relation: $O_{n-1} \to O_n$ is not reversible, so $O_{n-1}$ and $O_n$ are not equivalent under the transition structure. Novelty ($\Delta_n \neq 0$) ensures $O_n \neq O_{n-1}$, so the order is non-trivial. The composition of transitions defines transitivity. Together: asymmetry, irreflexivity, and transitivity give a strict partial order compatible with the chain structure. □

**Remark 4.3** (Time as derived structure)**.** *The arrow of time is not postulated. It is the oriented order generated by the chain of non-injective transitions under A2. The index n acquires its temporal meaning from this order — not the reverse. The quantitative structure of this projective temporal ordering — cumulative projected capacity and its numerical validation from the spectral admissibility cascade — is developed in [14].*

**Remark 4.4** (Admissible transitions as a state machine)**.** *The admissible neighbourhood* $\mathcal{A}(O_{n-1})$ *(Remark 2.2) gives the chain* (2) *the structure of an oriented state machine: nodes are admissible projections, edges are admissible transitions, and a physical history is a directed path in this graph. When* $|\mathcal{A}(O_{n-1})| = 1$*, the successor is uniquely determined by the admissibility structure alone — no external law is required. When* $|\mathcal{A}(O_{n-1})| > 1$*, branching occurs; the relative admissibility weights of the branches are the projective analogues of quantum probabilities (Section 5). In both cases, no temporal parameter external to the transition structure is needed: the index* $n$ *acquires its temporal meaning entirely from the orientation of the admissibility graph.* Dynamics is the oriented topology of the admissibility graph, not a law imposed upon a pre-existing time.

**Remark 4.5** (Observable reality as projective position)**.** *The admissible neighbourhood* $\mathcal{A}(O_{n-1})$ *is not a fixed space of possibilities. It shifts at every step, because* $O_n$ *conditions* $\mathcal{A}(O_n)$*, which differs generically from* $\mathcal{A}(O_{n-1})$*: the projection following step* $n$ *is never performed from an invariant reference frame. Observable reality is therefore not a static projection of a fixed substrate onto a fixed observable space. It is the* current projective position within the space of admissible reprojections*: the trajectory* $O_0 \rightsquigarrow O_1 \rightsquigarrow \cdots \rightsquigarrow O_n$ *is constitutive of what is real at step* $n$*, not contingent upon it. The cumulative index* $I(n)$ *measures not elapsed time but accumulated projective divergence from the initial position — how far the sequence has moved through the space of admissible neighbourhoods. In particular, the projection does not reveal a pre-existing reality: it redefines the context from which the next reality may appear.*

### 4.4 Projective incompleteness

Theorem 5.7 establishes that the admissibility structure forces $[X, \sigma(X)] = Z \neq 0$. This non-commutativity has an immediate structural consequence that is logically prior to — and independent of — the wave and phase structure of Section 5.

**Corollary 4.6** (Projective incompleteness)**.** *Under A1–A3 and Born–Infeld parity, no admissible projection* $\Pi_n$ *resolves* $\Omega_n^{(c)}$ *completely. Every admissible projection has a non-trivial kernel and leaves a strictly positive informational residue:*

$$\ker \Pi_n \;\neq\; \{0\}, \qquad S_{\Pi_n} \;>\; 0, \tag{12}$$

*for every admissible* $\Pi_n$*.*

*Proof.* By Theorem 5.7, $[X, \sigma(X)] = Z \neq 0$. A projection $\Pi_n$ with $\ker \Pi_n = \{0\}$ would simultaneously resolve both $X$ and $\sigma(X)$ without residual ambiguity, requiring their simultaneous diagonalisation and hence $[X, \sigma(X)] = 0$.

This contradicts $Z \neq 0$. Therefore $\ker \Pi_n \neq \{0\}$, and the projection entropy $S_{\Pi_n} > 0$ follows from Proposition 4.1. □

**Remark 4.7** (Structural fluctuations as necessary, not contingent)**.** *Corollary 4.6 changes the epistemic status of the structural fluctuations introduced in Section 4.2. They are not merely a consequence of abstract non-injectivity (A2 alone), but a* necessary consequence of the non-commutativity forced by A1–A3*: any projection that left no residue would require* $[X, \sigma(X)] = 0$*, contradicting Theorem 5.7. The quantum vacuum is therefore not a state from which all fluctuations have been removed; it is the state from which complete removal is structurally forbidden.*

**Remark 4.8** (Projective incompleteness and the first transition)**.** *Corollary 4.6 also resolves a potential question about the origin of the first admissible transition: what initiates* $\Pi_1$ *from* $\Pi_0$*? A perfectly self-closed initial state — one with* $\ker \Pi_0 = \{0\}$ *— is structurally forbidden. The initial projection necessarily leaves a non-zero residue, which constitutes the first difference and guarantees that* $\mathcal{A}(O_0)$ *is non-empty. No external mechanism is required: the first transition is a structural necessity, not a contingent event. This is the projective analogue of the impossibility of a perfectly symmetric initial state: the admissibility constraints that select the Heisenberg structure also ensure that a featureless origin is inadmissible.*

# 5 Admissibility, Wave, and Phase Coherence

## 5.1 The admissibility principle

Axiom A3 introduces a physical constraint on transitions:

> A transition is admissible if and only if it preserves the full multiplicity of open admissible directions without selecting among them prematurely.

Premature selection means making a determinative choice that nothing in the current state $O_{n-1}$ justifies. Admissibility forbids it: the transition must remain structurally neutral among all open directions until projection locking (A4) forces resolution.

## 5.2 The proto-state

**Definition 5.1** (Proto-state)**.** *The* proto-state $\Psi_n$ *associated with a transition from* $O_{n-1}$ *to* $O_n$ *is the unresolved configuration shared between* $O_{n-1}$ *and* $O_n$*: the carrier of the transition, belonging fully to neither endpoint.*

The proto-state is physically real — it is the configuration of the system during the transition, before projection locking. It is not an epistemic representation of uncertainty about a definite state: it is the actual structural state of the system between two realized observable states.

**Proposition 5.2** (Structural indeterminacy of the proto-state)**.** *Under A1 and A3, the proto-state $\Psi_n$ is genuinely indeterminate among the admissible directions in $F_n$.*

*Proof.* By A1, $F_n$ is non-empty. By A3, the transition does not select prematurely among the elements of $F_n$. Therefore $\Psi_n$ is not identified with any particular element of $F_n$ before projection locking: it is open across all of $F_n$ simultaneously. This is structural indeterminacy, not epistemic ignorance. □

### 5.3 Wave, phase, and coherence

The proto-state, being structurally open across all elements of $F_n$ simultaneously, propagates through all admissible directions at once. This is the physical origin of wave-like behaviour: not a metaphor, but a direct consequence of admissibility.

A configuration propagating through multiple directions simultaneously carries relative phase structure between those directions. Phase is therefore not postulated — it is a consequence of structural indeterminacy.

Admissibility (A3) preserves this phase structure: premature selection would collapse the relative phases, but A3 forbids it. The chain of consequences is therefore:

$$\begin{aligned} \text{A3} &\implies \text{structural indeterminacy} \\ &\implies \text{proto-state (wave)} \\ &\implies \text{phase} \\ &\implies \text{coherence preserved.} \end{aligned} \tag{13}$$

**Remark 5.3** (The admissible fibre as precursor of Hilbert structure)**.** *The fibre $F_n$, equipped with the phase structure forced by A3, is the natural precursor of an effective Hilbert sector: it carries complex amplitudes and interference structure. This is not yet a full Hilbert space — the inner product, completeness, and probability measure require additional derivation (see Q1, Q2 in Section 7). But the amplitude structure and interference follow from A3 alone.*

**Remark 5.4** (Double-slit experiment)**.** *In the double-slit experiment, the proto-state propagates through both slits because the admissibility constraints do not select between them: both are open directions in $F_n$. The interference pattern is a direct consequence of coherent propagation through all admissible* *directions. The impact on the screen is the moment the proto-state reaches the Born–Infeld saturation threshold — projection locking (A4) forces it onto an available shell, resolving it into a localized state $O_n$.*

### 5.4 Non-factorisability and the emergence of Heisenberg structure

Axiom A3 does not only enforce phase coherence. It also constrains the algebraic structure of the admissible fibre $F_n$ itself, forcing it to carry a Heisenberg structure rather than an arbitrary unitary symmetry.

**Remark 5.5** (The structural tension between A2 and A4)**.** *The non-factorisability established in this section can be understood as the minimal consistent resolution of a structural tension between A2 and A4.*

*A2 asserts that the multiplicity of admissible configurations in $F_n$ cannot be eliminated: distinct configurations lead to the same observable, and this indistinguishability is structural, not epistemic. A4 asserts that the observable outcome is unique at locking. Together, they impose an irresolvable constraint: the multiplicity cannot disappear (A2 would be violated), yet it cannot remain directly observable (A4 would be violated).*

*A3 — admissibility as non-premature selection — is the operative axiom that determines how this tension resolves. It forbids any factorisation of $F_n$ that would make the multiplicity distinguishable prior to locking, forcing it to organise into a globally coherent, irreducible structure. Lemma 5.6 formalises this constraint; Theorem 5.7 identifies the resulting structure as Heisenberg type. Quantum structure is therefore the minimal algebraic form compatible with the joint constraint of A2, A3, and A4.*

**Lemma 5.6** (Admissible non-factorisability)**.** *Under A1–A3, any decomposition*

$$F_n \;=\; F^{(1)} \oplus F^{(2)}$$

*that is stable under the admissible generators of $F_n$ is forbidden prior to projection locking. In particular, A3 forces the action of admissible generators on $F_n$ to be irreducible.*

*Proof.* A3 forbids premature selection among open directions. A decomposition of $F_n$ stable under admissible generators separates directions by invariants that are internal to the fibre — invariants carried by the generators themselves, which are constitutive of the admissibility structure by A1. Such a decomposition therefore resolves the proto-state partially, by assigning each open direction to a definite sector, prior to any Born–Infeld saturation event. This contradicts A3. Irreducibility follows immediately: any proper stable sub-representation would yield such a forbidden decomposition. □

**Theorem 5.7** (A1–A3 and BI parity force Heisenberg structure)**.** *Under A1–A3 and Born–Infeld parity, the symmetry group of the admissible fibre $F_n$ is isomorphic to* $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ *for some prime $q$. Consequently, $F_n$ is isomorphic to the Weil representation $V_\rho$ on $L^2(\mathbb{Z}/q\mathbb{Z})$.*

*Proof sketch.* Let $X$ and $\sigma(X)$ be the conjugate generators of $F_n$ under Born–Infeld parity. If $[X, \sigma(X)] = 0$, then $F_n$ decomposes into simultaneous eigenspaces, yielding a stable decomposition forbidden by Lemma 5.6. Hence $[X, \sigma(X)] \neq 0$. Minimality (A1) forces the commutator $Z := [X, \sigma(X)]$ to be central: any non-central $Z$ would extend the algebra beyond the conjugate pair, contradicting the minimal generative closure forced by A1. The group $\langle X, \sigma(X)\rangle$ is then a non-trivial central extension of $(\mathbb{Z}/q\mathbb{Z})^2$ by $\langle Z\rangle$, isomorphic to $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ by the classification of such extensions. Lemma 5.6 further implies that $|Z|$ is prime: if $|Z|$ were composite, the representation of $F_n$ would decompose over the characters of $\langle Z\rangle$, yielding a forbidden stable decomposition. Finally, Born–Infeld parity forces $c \neq 0$ [3], so the central character is non-trivial; the discrete Stone–von Neumann theorem then identifies $F_n \simeq V_\rho$ uniquely. The detailed proof of each step is given in the companion paper [12]. $\square$

$$\begin{aligned}
&\text{A3} && \Longrightarrow \text{ no admissible factorisation } \Longrightarrow [X, \sigma(X)] \neq 0 \Longrightarrow \text{ central extension,}\\
&\text{A1} && \Longrightarrow \text{ minimal generative closure } \Longrightarrow \text{ Heisenberg algebra,}\\
&\text{A3} && \Longrightarrow \text{ irreducibility } \Longrightarrow |Z| = q \text{ prime } \Longrightarrow \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z}),\\
&\text{BI parity} && \Longrightarrow c \neq 0 \Longrightarrow F_n \simeq V_\rho.
\end{aligned} \tag{14}$$

**Remark 5.8** (Uncertainty principle as a corollary)**.** *Theorem 5.7 subsumes the open direction Q6 of Section 7. The incompatibility of conjugate observables $X$ and $\sigma(X)$ — i.e. the Heisenberg uncertainty principle — is a direct corollary: it follows from $[X, \sigma(X)] \neq 0$ together with the standard operator inequality. The uncertainty principle is therefore not a property of the observables, but a property of the non-resolved proto-state: it is the algebraic signature of the prohibition on premature selection encoded in A3.*

**Remark 5.9** (Canonical quantisation as structural necessity)**.** *The relation $[X, \sigma(X)] = Z$ defines a minimal discrete symplectic structure, of which the Heisenberg algebra is the canonical quantisation. Theorem 5.7 therefore establishes that canonical quantisation is not a modelling choice: it is the unique algebraic structure compatible with the requirement that the proto-state remain genuinely non-resolved until projection locking.*

A quantum superposition is an unresolved proto-state: a physically real configuration whose fibre maintains coherent relations between multiple open admissible directions. There is no ontological paradox — superposition is the normal condition of a system between two transitions.

A measurement is a transition whose admissibility constraints are strong enough to force resolution of the proto-state onto a single admissible direction. Free evolution and measurement differ in the strength of their admissibility constraints, not in their fundamental mechanism.

In this sense, quantum mechanics is the effective theory of physically real unresolved admissible structures and their resolution.

### 5.5 The wavefunction as effective representation of the proto-state

A quantum wave is not a physical substance propagating in space. It is the effective representation of a physically real unresolved proto-state. It encodes the coherent structure of admissible directions prior to projection locking.

More precisely: the proto-state $\Psi_n$ is real; the wavefunction $\psi$ is a coordinate description of the coherent admissible structure carried by $\Psi_n$ in the spectral realisation $V_\rho$ of $F_n$. When we write $\psi$, we are not describing a substance propagating in space — we are encoding the relational phase structure of an unresolved configuration in the space of admissible directions. Quantum mechanics does not describe realized physical states: it describes states in the process of being resolved.

This resolves the standard interpretational difficulties without additional postulate:

- *Superposition* is the normal condition of a proto-state — the simultaneous structural openness across all elements of $F_n$.
- *Interference* is the direct consequence of coherent propagation through multiple admissible directions before resolution.
- *Collapse* is not a physical process applied to a pre-existing state but the resolution of the proto-state at projection locking — the transition from unresolved admissible structure to a realized observable outcome.

### 5.6 Why complex amplitudes

Given that the proto-state is a physically real, unresolved, coherent configuration, what is the minimal mathematical structure capable of representing it?

Real positive weights are insufficient. They describe ignorance about which of several pre-existing alternatives is realized — a classical mixture. The proto-state is not a mixture: it is a single coherent configuration with definite phase relations between admissible directions.

Real signed weights are also insufficient. They allow opposition of sign but do not support a rich enough phase structure under composition. In particular, they cannot represent the full coherence of a proto-state whose directions combine under the SU(2) structure forced by O23.

The minimal structure that can: (i) represent simultaneous coherent openness across multiple directions, (ii) encode their relative phase relations, (iii) compose correctly under the SU(2) symmetry of the admissible fibre, (iv) support interference — the physical consequence of coherence before resolution — is a complex amplitude.

Therefore: complex amplitudes are not introduced by convention. They are the minimal representation of a coherent unresolved proto-state. The imaginary unit encodes the relative phase between admissible directions; it enters because the proto-state is not a mixture but a coherent superposition.

### 5.7 Why the Born weight is the squared norm

Upon projection locking (A4), the unresolved proto-state is resolved onto a single admissible direction. The quantity associated with each direction at resolution — the *intensity of realisation* — must satisfy four structural requirements:

(i) *Positivity*: a weight on admissible directions cannot be negative.

(ii) *Global phase invariance*: multiplying $\psi$ by a global phase $e^{i\theta}$ does not change the physical proto-state, so the weight must be invariant under this transformation.

(iii) *Normalisability*: the total weight over all admissible directions must be finite and summable to one.

(iv) *Coherence preservation before resolution*: the weight must not erase phase relations prematurely — interference must remain possible up to the moment of resolution.

The squared norm $|\psi|^2$ is the unique minimal quantity satisfying all four requirements. The norm $|\psi|$ fails (ii) under phase shifts and (iv) — it introduces real-valued interference which is not the correct structure. Any higher power $|\psi|^p$ for $p \neq 2$ fails (iii) under composition of superpositions. Any non-analytic function of $\psi$ fails (iv).

The Born weight is therefore not a postulate. It is the positive intensity associated with the resolution of a coherent admissible structure — the unique minimal such intensity compatible with the four structural requirements.

**Remark 5.10** (Relation to the Born rule derivation in the companion paper)**.** *The argument above establishes the structural necessity of the squared* *norm as the resolution intensity of the proto-state. The companion paper [8] makes this precise as a theorem in the SU(2) parity sector: the Born rule $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi\rangle|^2$ is the unique probability assignment compatible with the coherent fibre structure, the SU(2) observable algebra, and the derived singlet correlator. The two derivations are complementary: the present argument gives the structural motivation; the companion gives the uniqueness proof.*

**Remark 5.11** (The wavefunction lives in the admissible fibre, not in space)**.** *A common source of confusion is to picture the wavefunction as an object propagating in physical space. In the present framework, $\psi$ encodes the structure of $F_n$ — the admissible directions from $O_{n-1}$. Its concrete realisation is the spectral sector $V_\rho$ induced by the admissibility constraints on the Heisenberg Cayley graph (O-series). $F_n$ and $V_\rho$ are not pre-existing spaces: they are the spaces defined by the admissibility constraints. The wavefunction does not propagate in space; it encodes the relational phase structure of an unresolved admissible configuration.*

Axiom A4 establishes that resolution is discrete. This discreteness is not postulated: it follows from the interaction of two structures already present in the framework.

### 5.8 Two types of discreteness

The Born–Infeld saturation condition [3] operates on a continuous amplitude function. Its saturation locus $\mathcal{L}_{\mathrm{BI}} \subset \mathbb{R}$ is generically real-valued — its elements carry no information about shell structure.

The observable space decomposes discretely:

$$O = \bigsqcup_{n\in\mathbb{N}_q} O_n, \tag{15}$$

indexed by BFS shells of the Heisenberg Cayley graph [18]. No component $O_{n'}$ exists for non-admissible depths.

The proto-state is resolved at the first element of $\mathcal{L}_{\mathrm{BI}} \cap \mathbb{N}_q$ — the first depth where Born–Infeld saturation coincides with an available shell. This is the Projection Locking theorem [18].

**Theorem 5.12** (Discreteness from projection locking, after O22)**.** *Under A3 and A4, quantum transitions are discrete. The discreteness is a consequence of the interaction of continuous Born–Infeld saturation with the discrete admissibility filter of $\Pi_n$, not a postulate.*

### 5.9 Three stable directions

The number of independent stable admissible directions in the fibre is exactly three, derived algebraically from the admissibility constraints alone [21]:

$$\begin{aligned}
\text{BI parity} &\Rightarrow \text{fibre involution } c \leftrightarrow q - c \\
&\Rightarrow \text{non-abelian admissible support} \\
&\Rightarrow \mathbb{H}\text{-minimality} \\
&\Rightarrow \dim_{\mathbb{R}} \operatorname{Im} \mathbb{H} = 3 \\
&\Rightarrow \Sigma_c(n_3) = 3.
\end{aligned} \tag{16}$$

The emergence of exactly three stable directions — and with it, three particle generations — is algebraic in origin, not geometric or phenomenological.

# 6 Entanglement and Bell Non-Applicability

## 6.1 Entanglement as shared ancestral fibre

Two observable systems $A$ and $B$ are entangled when they share a common ancestral fibre: a past state $O_{n-k}$ in which they formed a single non-factorisable configuration.

The shared fibre $F_{n-k}$ reduces the admissible directions for each subsystem: the admissible directions for $A$ are constrained by $B$, and vice versa. The correlations arise not from any interaction at the moment of measurement, but from the persistence of the shared fibre — a residue of the past transition in which $A$ and $B$ were joined.

## 6.2 Why Bell's factorisation assumptions do not apply

Bell's theorem [28] constrains models in which measurement outcomes are determined by a variable $\lambda$ satisfying: (i) $\lambda$ is local (assignable separately to $A$ and $B$), (ii) $\lambda$ is present (defined at the moment of measurement), (iii) $\lambda$ is independent (statistically independent of measurement settings).

The fibre variable in the present framework satisfies none of these conditions. It is not local (it belongs to the shared fibre, not to a spatial region), not present (it belongs to a past transition, not the current state), and not independent ($A$ and $B$ share the same ancestral fibre, which cannot be factorised).

The Bell factorisation assumption $P(a, b|\lambda) = P(a|\lambda) \cdot P(b|\lambda)$ fails because it requires $\lambda$ to be assignable separately to $A$ and $B$: the shared fibre is structurally non-factorisable. Bell-type inequalities are therefore not violated in this framework — their preconditions do not hold. This is established formally in the companion paper [1].

## 7 Summary: What Is Derived from What

Table 1 summarises the derivation chain.

### Partial spectral stratification: an organising pattern from the companion programme

The four axioms A1–A4 are the axiomatic primitive of the programme. They do not by themselves determine the interaction content of the emergent theory; that content emerges from the spectral structure of the admissible operator, developed in the Q-series companion papers. The companion papers [10, 27, 25] establish the following partial pattern, which organises the physical interpretation of the Seeley–DeWitt expansion of $S_\Pi[g, A]$:

**Principle 7.1** (Spectral stratification of effective interactions)**.** *The physical role of an emergent interaction is determined not only by the symmetry it carries, but by the spectral order at which the admissible operator responds to variation. The gravitational sector appears at the first non-trivial horizontal response; the gauge sector appears at the vertical curvature response; and mixed gauge–gravity terms appear at the next coupled order.*

The evidence base for Principle 7.1 is as follows:

| Spectral order | Sector | Geometric origin |
|---|---|---|
| $a_2$ | gravity | horizontal variation of the base metric [10] |
| $a_4$ | gauge | vertical variation of the fibre connection [27] |
| $a_6$ | mixed coupling | joint base–fibre curvature terms [25] |
| $a_{\rm spin}$ | fermionic structure | Dirac-type admissible operator (open) |

**Remark 7.2.** *Principle 7.1 is* not *derived from A1–A4 alone. Its correct epistemic status is: established by the companion programme, consistent with A1–A4, and subject to the conditionalities of each companion paper. In particular, the identification* $G_\Pi = \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ *remains conditional on* [H-color] *[26, 24], the Lorentzian continuation of the spectral functional remains an open step, and the fermionic entry is a structural prediction, not a derived result. The test of this prediction is whether admissibility forces a Clifford module and a* $\mathbb{Z}_2$*-grading on the admissible operator.*

### Open directions

**Q2 — Born rule beyond SU(2).** The Born rule $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi\rangle|^2$ is derived in [8] for the SU(2) parity sector. A first step beyond this sector is taken in [15], which derives the Born rule, singlet correlator, and Tsirelson bound for the co-admissible spin-$\frac{3}{2}$ sector on the binary icosahedral group $2I$, and identifies SU(2) as the unique stable sector under admissibility refinement in the large-$p$ limit. The full extension to arbitrary observables, multipartite systems, and continuous spectra requires extending the fibre structure beyond the two-element parity case and remains the primary open direction.

**O31/O32 — Colour sector and recursive non-injectivity.** The SU(2) derivation proceeds via conjugate pairs $\{c, q-c\}$. For primes $q \equiv 1 \pmod 3$, the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ admits colour triplets $\{c_1, c_2, c_3\}$ satisfying $c_1+c_2+c_3 \equiv 0 \pmod q$ and pairwise non-conjugacy. Under Hypothesis [H-color] (triplet co-admissibility, numerically supported in O32 [24]), O31 [26] derives SU(3) as the admissibility fixed point of the colour fibre, completing the identification $G_{\mathrm{SM}} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

Each element $c_i$ of a valid triplet already carries spin-$\frac{1}{2}$ (from its conjugate pair $\{c_i, q - c_i\}$, O29 [19]) and one colour charge (from the triplet structure). This is the minimal quantum-number content of a quark. The identification of quarks as stable projected configurations in $\mathrm{Im}\,\Pi$ carrying these quantum numbers, and the derivation of colour confinement as a structural consequence of the projection $\Pi_{\mathrm{colour}} : \{c_i\} \to \{\text{colour-neutral objects}\}$ being non-injective (ENI [13], Corollary 6), are the primary open directions of the colour sector.

More broadly, ENI Corollary 6 establishes that the four axioms force non-injectivity not only at the level of the fundamental projection $\Pi_1$, but at every stratum of a tower of genuinely distinct emergent levels. The programme's current architecture instantiates two strata: the pair level (leptons, $\Pi_1$) and the triplet level (colour sector, $\Pi_2$). The structural derivation of the full particle hierarchy from this recursive tower is the central open direction of the programme.

**Spectral quasi-fixed points and stable structures.** The admissible neighbourhood $\mathcal{A}(O_n)$ varies generically with $n$ (Remark 4.5). However, configurations $O^*$ for which $\mathcal{A}(O^*)$ is approximately self-similar — in the sense that projection from $O^*$ tends to return to a spectrally close neighbourhood — define *spectral quasi-fixed points* of the admissible flow. These are not static states, which would contradict the irreversibility established in Proposition 4.1, but projective orbits that traverse the same spectral region repeatedly. Stable physical structures — particles, gauge symmetries, and coupling constants — are candidates for identification with such quasi-fixed points: their observed stability would then be a projective property, arising from the near-invariance of their admissible neighbourhood under repeated projection, not from any ontological fixity. The universality of $a_q(s) \cdot q^2 \approx 2$ across all tested primes (O-series numerics) is consistent with the existence of such an attractor in the admissible spectral flow, independent of $q$. This numerical observation acquires a structural anchor from the companion paper

Q10 [23]: the horizontal coefficient $A_H(q)$ of the effective spatial co-metric satisfies $A_H(q) \to 2$ as $q \to \infty$, where the limiting value is identified as the Casimir eigenvalue

$$C_{\mathrm{su}(2)}\big|_{j=1} \;=\; j(j+1) \;=\; 2 \tag{17}$$

of the spin-1 representation $\mathrm{Sym}^2(V_\rho)$ — the unique su(2)-invariant quadratic form on the effective spatial sector (Q10, Proposition 5.1, via Schur's lemma). The epistemic chain leading to this identification is:

$$[X,Y] \neq 0 \;\Rightarrow\; \text{no perfect resolution} \;\Rightarrow\; [\mathrm{U}] \;\Rightarrow\; \mathrm{su}(2)\text{-isotropy} \;\Rightarrow\; C_{\mathrm{su}(2)}|_{j=1} = 2 \;\Rightarrow\; A_H(q) \to 2, \tag{18}$$

where: the first implication is Corollary 4.6; the universality hypothesis [U] is structurally supported and numerically confirmed for $q \le 211$ [20, 23]; the su(2)-isotropy step is Q10 Lemma 4.1; and the Casimir identification is Q10 Proposition 5.1 (unconditional). The convergence $A_H(q) \to 2$ is established conditionally on [U] in Q10 Theorem 1.1. The attractor value 2 is therefore not a free parameter but is forced by the representation theory of the admissible fibre, providing a representation-theoretic interpretation of the quasi-fixed point. Formalising the existence and classification of quasi-fixed points of $\mathcal{A}$ is an open direction of the programme.

Theorem 5.7 establishes $F_n \simeq V_\rho$ at finite prime $q$. The relationship between this discrete structure and continuous spacetime geometry in the large-$q$ limit — i.e. how $V_\rho$ on $L^2(\mathbb{Z}/q\mathbb{Z})$ approaches $L^2(\mathbb{R})$ and how the BFS shell structure produces a four-dimensional effective geometry — is addressed in the companion spectral geometry paper [11] (Section 3.5). That paper reduces Q5 to a single spectral tightness condition (Mosco tightness of the admissibility forms), confirms this condition quantitatively across all tested primes, and proves the embedding in the low-frequency regime relevant to admissible sequences. The identification of the effective metric from the principal symbol of the limit operator $L_\Pi$ and the Lorentzian signature selection are established in the companion paper Q5b [2].

**Q14 — Fermionic structure and chirality.** The spectral stratification of Principle 7.1 identifies the next open level of the programme: the fermionic sector. The companion papers Q12 and Q13 [27, 25] derive bosonic gauge and gravitational dynamics from a Laplace-type admissible operator $\mathcal{A}_{g,A}$. Fermionic structure requires a Dirac-type operator $\mathcal{D}_{g,A}$ satisfying $\mathcal{D}^2 = \mathcal{A}_{g,A} + \frac{1}{4}R$ (Lichnerowicz formula), whose $\mathbb{Z}_2$-grading encodes chirality. The central open question is whether the admissibility structure forces a Clifford module structure on the admissible fibre and thereby selects a $\mathbb{Z}_2$-grading: if so, the chiral asymmetry of the Standard Model becomes a structural consequence of admissibility; if not, this constitutes a structural boundary of the programme.

| Concept | Source | Status |
|---|---|---|
| *Direct axiomatic consequences (A1–A4)* | | |
| Irreversibility | Non-injectivity (A2) | Proved (Prop. 4.1) |
| Temporal order | Irreversibility + novelty (A2) | Proved (Thm. 4.2) |
| Proto-state (wave) | Unresolved multiplicity (A1, A3) | Derived (Prop. 5.2) |
| Phase coherence | Admissibility preserves open directions (A3) | Derived |
| Discrete transitions | Projection locking (A4, O22) | Proved (Thm. 5.12) |
| Three stable directions | $\dim_{\mathbb{R}} \operatorname{Im} \mathbb{H} = 3$ (O23) | Proved |
| Entanglement | Shared non-factorisable ancestral fibre | Derived |
| Bell non-applicability | Non-local, non-independent fibre | Proved [1] |
| Born rule (SU(2) sector) | Correlator + coherence [8] | Proved |
| $F_n \simeq V_\rho$ (Heisenberg structure) | Non-factorisability (A1, A3) + BI parity | Proved (Thm. 5.7) |
| *Key results of the companion programme (see [17] for the full derivation chain)* | | |
| SU(2) as admissibility fixed point | Spectral convergence (Q2) [15] | Proved |
| Effective spacetime geometry, $D_{\text{hom}} = 4$ | BFS stratification (Q5b, Q6b) [2, 5] | Proved |
| Einstein equation $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ | Spectral entropy variation [10] | Proved (structural) |
| SU(2) gauge structure | Quaternionic rigidity + Q6a [16, 9] | Proved |
| Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ | Spectral entropy $a_4$ variation (Q12) [27] | Proved (structural) |
| Gauge–matter coupling $G_{\mu\nu} = 8\pi G_N T^{YM}_{\mu\nu}$ | Joint variation $S_\Pi[g, A]$ (Q13) [25] | Proved (structural) |
| Hierarchy $G_N g^2_{YM} \ll 1$ without fine-tuning | Seeley–DeWitt stratification (Q13) [25] | Structural |
| Tower of non-injectivity | ENI Corollary 6 [13] | Proved |
| *Conditional on named hypothesis* | | |
| $G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ | Colour co-admissibility (O31, O32) [26] | Conditional on [H-color] |
| *Open* | | |
| Full Hilbert structure (continuous spectrum) | Beyond discrete fibre; large-$q$ limit | **Open** |
| Born rule (general observables) | Beyond $\text{SU}(2)/V_\rho$ sector | **Open** |
| Colour charges, quark sector | Individual sectors $c_i$ (O31, O32) | **Open** |

Table 1: Derivation chain from the four axioms. "Derived": structural consequence of the axioms. "Proved": explicit proof in this paper or a cited companion. "Proved (structural)": established via structural argument internally consistent with the framework; full formalisation deferred. "Conditional": proved subject to a named hypothesis.

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